

Data Processing Code Breakdown: Functionality and Power Management

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1 System Overview

The dog tracker is an ESP32-S3-based device designed for monitoring a dog's activity, location, and health, powered by a 3.7 V, 350 mAh (1.295 Wh) LiPo battery. It aims to achieve a battery life of approximately 90+ days with a daily power consumption of 3.06 mAh. The system integrates the following components:

- **Platform:** ESP32-S3 microcontroller.
- **Interfaces:**
 - **BLE:** Primary interface for receiving GPS coordinates from a mobile app, active during motion or on-demand (2.5 hr/day, 0.3 mA, 0.75 mAh/day).
 - **Wi-Fi:** Used for OTA firmware updates and periodic log uploads (10 s/day, 70 mA, 0.20 mAh/day).
- **Sensor:** QMI8658 accelerometer/gyroscope for activity tracking and epilepsy detection (17 hr/day, 0.1 mA average, 1.70 mAh/day).
- **Power Modes:**
 - Light sleep: After 5–10 min inactivity, 5 hr/day, 0.05 mA, 0.25 mAh/day.
 - Deep sleep: After 10+ min inactivity or overnight, 6 hr/day, 0.02 mA, 0.12 mAh/day.
- **Other:** Epilepsy monitoring (10 s/day, 15 mA, 0.04 mAh/day), BLE idle (< 0.01 mAh/day).

The total daily consumption is 3.06 mAh, yielding a battery life of $\frac{350}{3.06} \approx 114.4$ days, exceeding the target of 45+ days.

2 Code Structure and Initialization

The `app_main` function initializes the system at boot or reset:

- **NVS Flash:** Initializes non-volatile storage for persistent data (e.g., BLE settings).
- **SPIFFS:** Sets up local storage for logs (`/spiffs/log.txt`).
- **Wi-Fi:** Initializes in station mode (not connected yet).
- **BLE:** Initializes NimBLE with device name “MJ”, power level `ESP_PWR_LVL_P9`, and bonding security. Creates a service (UUID: `180A`) with characteristics:
 - `RX_CHAR_UUID` (write): Receives GPS coordinates.
 - `ECHO_CHAR_UUID` (read/notify): Echoes coordinates.
 - `DIST_CHAR_UUID` (read/notify): Reports distance (meters).
 - `SPEED_CHAR_UUID` (read/notify): Reports speed (m/s).
 - `STEPS_CHAR_UUID` (read/notify): Reports step count.

- INFO_CHAR_UUID (write): Receives dog info (breed, age, size).
- ACTIVITY_CHAR_UUID (read/notify): Reports activity (Resting, Walking, Running).
- BATTERY_CHAR_UUID (read/notify): Reports battery life (days).
- **BLE Advertising:** Starts with 100 ms interval, name "MJ", and service UUID.
- **Sensor:** Initializes QMI8658 via I2C (100 kHz, SDA: GPIO15, SCL: GPIO14) with accelerometer (4G, 1000 Hz) and gyroscope (64 DPS, 896.8 Hz).
- **Battery:** Sets batteryRemaining = 350 mAh and notifies initial battery life (~114.4 days).
- **Task:** Starts read_sensor_data FreeRTOS task for sensor sampling and power management.

Timing: Runs once at boot, taking seconds (negligible power impact).

3 BLE Operation

BLE handles GPS coordinate reception and dog info updates, with two modes:

- **Advertising:** Active when not connected, using low-duty-cycle (100 ms interval, 0.01 mAh/day).
- **Connected:**
 - **Trigger:** Mobile app connects (ServerCallbacks::onConnect).
 - **Actions:**
 - * Sets isBleActive = true and updates lastActivityTime.
 - * Receives GPS (lat, lon) via RX_CHAR_UUID (RxCallbacks::onWrite).
 - * Calculates:
 - Distance: Haversine formula (haversine_m).
 - Speed: Distance ÷ time since last update.
 - Steps: Distance ÷ stepSize (from dog breed, age, size).
 - * Notifies via ECHO, DIST, SPEED, STEPS characteristics.
 - * Logs to SPIFFS.
 - Receives dog info (breed, age, size) via INFO_CHAR_UUID (InfoCallbacks::onWrite) calculates stepSize, logs to SPIFFS.
 - **Disconnect:** Sets isBleActive = false, restarts advertising (ServerCallbacks::onDisconnect).

When Active:

- **Advertising:** Always when not connected.

- **Connected:** During motion (e.g., walks, 2.5 hr/day) or on-demand (e.g., app updates).
- **Scenarios:** Dog walks (GPS sent), user updates dog info.

When Off: Returns to advertising after app disconnects.

Conditions: Connected mode prevents sleep modes. Assumed 2.5 hr/day.

Power: $0.3 \text{ mA} \times 2.5 \text{ hr} = 0.75 \text{ mAh/day}$ (24.5% of 3.06 mAh).

4 Wi-Fi Operation

Wi-Fi is used for log uploads and potential OTA updates:

- **Initialization** (`wifi_init`): Sets up Wi-Fi in station mode.
- **Connection** (`wifi_connect`): Connects to SSID/password, sets `isWifiActive = true`.
- **Log Upload** (`wifi_upload_logs`):
 - Reads `/spiffs/log.txt`.
 - Sends logs via HTTP POST to server.
 - Clears logs after upload.
 - Disconnects Wi-Fi, sets `isWifiActive = false`.

When Active: Once daily for 10 s (e.g., 2 AM) to upload logs.

When Off: After 10 s upload or timeout.

Conditions: Activates if logs exist and 24 hr since last upload.

Power: $70 \text{ mA} \times 10 \text{ s} = 0.20 \text{ mAh/day}$ (6.5%).

5 Sensor Operation (QMI8658)

The QMI8658 sensor tracks activity and detects epilepsy events:

- **Initialization** (`setup_sensor`): Configures I2C, accelerometer (4G, 1000 Hz), gyroscope (64 DPS, 896.8 Hz).
- **Sampling** (`read_sensor_data`):
 - Every 10 s (`SENSOR_SAMPLE_INTERVAL`), active for 0.1 s (1% duty cycle).
 - Reads accelerometer (`acc.x`, `acc.y`, `acc.z`).
 - Calculates magnitude: $\sqrt{x^2 + y^2 + z^2}$, adjusts for gravity (`fabs(mag - 1.0)`).
 - Stores in moving average buffer (size 10).
 - Determines activity:

- * < 0.1 g: Resting
- * 0.1–0.5 g: Walking
- * > 0.5 g: Running
- * > 1.5 g: Increments `epilepsyEvents`

– Notifies via `ACTIVITY_CHAR_UUID`, logs to SPIFFS.

When Active: Every 10 s for 0.1 s, 17 hr/day (daytime).

When Inactive: During light sleep (5 hr) or deep sleep (6 hr).

Conditions: Samples when data is ready. Assumed 17 hr/day.

Power: $10\text{ mA} \times 1\% = 0.1\text{ mA} \times 17\text{ hr} = 1.70\text{ mAh/day}$ (55.6%).

6 Sleep Modes

Sleep modes reduce power during inactivity:

- **Light Sleep** (`enter_light_sleep`):
 - After 5 min inactivity (`LIGHT_SLEEP_TIMEOUT`).
 - Uses `esp_light_sleep_start`, 5 min timer.
 - 0.05 mA, 5 hr/day, 0.25 mAh/day.
- **Deep Sleep** (`enter_deep_sleep`):
 - After 10 min inactivity (`DEEP_SLEEP_TIMEOUT`) or overnight.
 - Uses `esp_deep_sleep_start`, reboots on wake.
 - 0.02 mA, 6 hr/day, 0.12 mAh/day.

When Active:

- Light sleep: 5 min inactivity, 5 hr/day (e.g., daytime rests).
- Deep sleep: 10 min inactivity or overnight, 6 hr/day.

When Exit: Light sleep wakes on timer/interrupt; deep sleep reboots.

Conditions: Blocked if `isBleActive` or Wi-Fi active.

Power: Light: 0.25 mAh/day (8.2%); Deep: 0.12 mAh/day (3.9%).

7 Epilepsy Monitoring

High-intensity motion is detected for epilepsy monitoring:

- **Detection:** In `detect_activity`, if `magnitude > 1.5 g`, increments `epilepsyEvents`.
- **Logging:** Included in activity logs to SPIFFS.

- **Power:** $15 \text{ mA} \times 10 \text{ s/day} = 0.04 \text{ mAh/day}$.

When Active: During sensor sampling if acceleration $> 1.5 \text{ g}$.

Conditions: Assumed 10 s/day.

Power: 0.04 mAh/day (1.3%).

8 Battery Management

Battery life is tracked and reported:

- **Tracking:** batteryRemaining starts at 350 mAh. Deducts:
 - Sensor: $0.1 \text{ mA} \times 0.1 \text{ s/sample}$.
 - Wi-Fi: $70 \text{ mA} \times 10 \text{ s daily}$.
 - Epilepsy: $15 \text{ mA} \times 10 \text{ s daily}$.
 - Light sleep: $0.05 \text{ mA} \times 5 \text{ hr}$.
 - Deep sleep: $0.02 \text{ mA} \times 6 \text{ hr}$.
- **Reporting:** Notifies life (batteryRemaining / 3.06) via BATTERY_CHAR_UUID at boot and after updates.

When Updated: At boot, after samples, uploads, or sleep.

Power: Total 3.06 mAh/day, 114.4 days.

9 Data Logging (SPIFFS)

Logs store data locally for later upload:

- **Initialization:** spiffs_init sets up /spiffs/log.txt.
- **Logging:** Stores GPS, dog info, activity via log_data.
- **Upload:** wifi_upload_logs sends logs, clears file.

When Active: On GPS, info, or sensor updates; uploaded daily.

Power: Negligible (0.01–0.02 mAh/day).

10 Daily Timeline Example

- **00:00–06:00:** Deep sleep (0.12 mAh).
- **06:00–08:30:** BLE active (walk, 0.75 mAh), sensor sampling (0.1 mA).
- **08:30–14:30:** Sensor active (0.1 mA), BLE idle (0.01 mAh), epilepsy event (0.04 mAh).
- **14:30–19:30:** Light sleep (0.25 mAh), periodic sensor wakes.

- **02:00:** Wi-Fi upload (10 s, 0.20 mAh).
- **19:30–00:00:** Sensor active (0.1 mA).

Total sensor time: 17 hr. Total: 3.06 mAh/day.

11 Power Consumption Summary

Component	Avg Current	Duration	Daily Usage
Sensor + MCU	0.1 mA	17 hr	1.70 mAh
BLE (Connected)	0.3 mA	2.5 hr	0.75 mAh
Wi-Fi Burst	70 mA	10 s	0.20 mAh
Light Sleep	0.05 mA	5 hr	0.25 mAh
Deep Sleep	0.02 mA	6 hr	0.12 mAh
Epilepsy Events	15 mA	10 s	0.04 mAh
BLE Idle	Negligible	10 min	0.01 mAh
Total			3.06 mAh/day

Table 1: Daily power consumption.

Battery life: $\frac{350}{3.06} \approx 114.4$ days.

12 Conclusion

The code ensures a 114.4-day battery life by optimizing BLE (2.5 hr/day), Wi-Fi (10 s/day), sensor sampling (1% duty cycle, 17 hr/day), and sleep modes (11 hr/day). It logs data to SPIFFS and uploads via Wi-Fi, with epilepsy detection and battery tracking. Real-world testing is recommended to validate assumptions.